

Complex Analysis – 4

Вычеты: интегралы и суммирование рядов

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Теоретический минимум.

Опр. If f has an isolated singularity at a with Laurent expansion $\sum_{n \in \mathbb{Z}} c_n (z-a)^n$, then $\operatorname{res}_a f = c_{-1}$. For a pole of order m , $\operatorname{res}_a f = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} ((z-a)^m f)$, and for a simple pole of P/Q it is $P(a)/Q'(a)$.

Теорема 1 (теорема о вычетах) $\oint_\gamma f = 2\pi i \sum_a n(\gamma, a) \operatorname{res}_a f$ for a cycle γ null-homotopic in the domain and avoiding the singularities.

Теорема 2 (лемма Жордана) If $M(R) = \max_{|z|=R, \operatorname{Im} z \geq 0} |f(z)| \rightarrow 0$, then $\int_{C_R} f(z) e^{i\lambda z} dz \rightarrow 0$ for every $\lambda > 0$, where C_R is the upper semicircle.

Факт 1 Суммирование рядов: $\pi \cot \pi z$ has simple poles at all integers with residue 1, and $\pi / \sin \pi z$ has residue $(-1)^n$ at n ; integrating $f(z) \pi \cot \pi z$ over large squares turns $\sum_{n \in \mathbb{Z}} f(n)$ into a finite sum of residues of f .

1. Prove the residue theorem, and compute $\int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$ ($a > 1$) and $\int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^n}$.
2. Prove Jordan's lemma and deduce $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$; explain carefully how the indentation at the origin is handled.
3. Compute $\int_0^\infty \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin \pi a}$ for $0 < a < 1$ by the keyhole contour, and deduce the reflection formula $\Gamma(a)\Gamma(1-a) = \frac{\pi}{\sin \pi a}$.
4. (S.T. Yau CSMC 2017, team) Show that a single-valued analytic branch of $\sqrt{1-z^2}$ can be defined in any region such that the points ± 1 lie in the same connected component of the complement. What are the possible values of $\oint \frac{dz}{\sqrt{1-z^2}}$ over closed curves in such a region?
5. Prove that $\sum_{n \in \mathbb{Z}} \frac{1}{n^2 + a^2} = \frac{\pi}{a} \coth \pi a$ by integrating $\frac{\pi \cot \pi z}{z^2 + a^2}$ over expanding squares, and deduce $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$ by letting $a \rightarrow 0$.